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% Simulation of D.C. Generator Characteristics
% 1. Open Circuit Characteristics (OCC)
% 2. External Load Characteristics
%
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clear; clc; close all;

%% Part 1: Open Circuit Characteristics (OCC)
% Simulating the non-linear relationship between Field Current and
Generated Voltage

If = 0:0.1:5; % Field current (A) ranging from 0 to 5 Amps

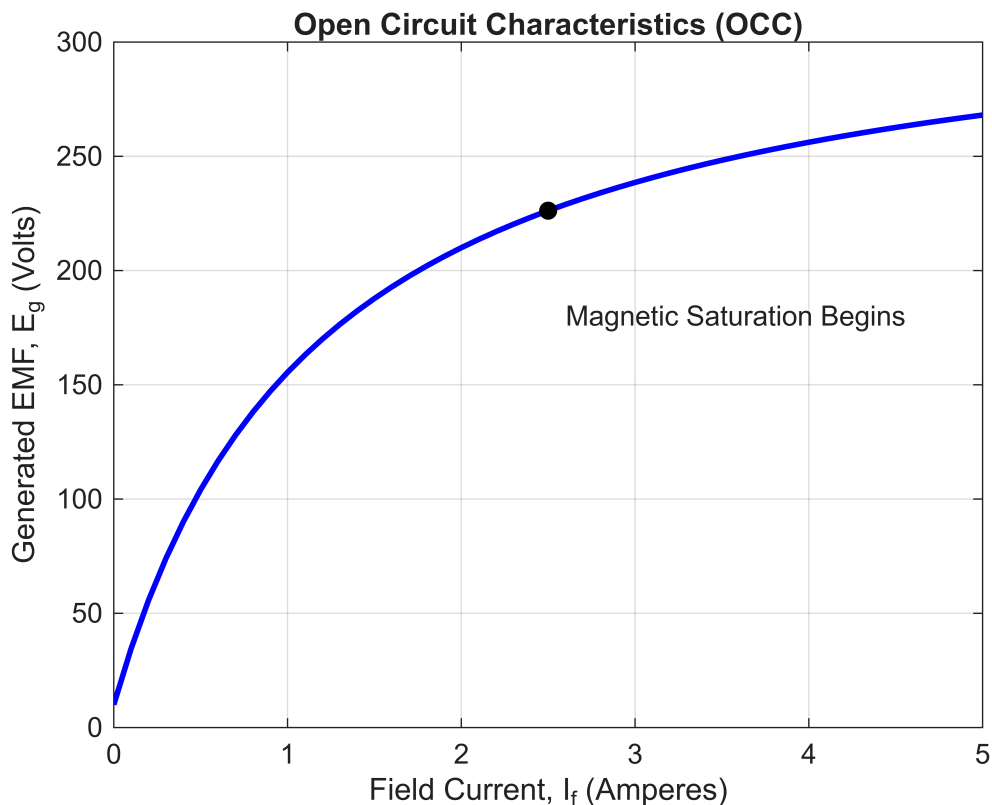
% Frölich equation parameters for magnetic saturation
a = 320; % Maximum saturation voltage limit
b = 1.2; % Curve shape parameter

% Calculate Generated EMF (Eg)
Eg = (a .* If) ./ (b + If);

% Add a small amount of residual magnetism (voltage present when If = 0)
residual_voltage = 10;
Eg = Eg + residual_voltage;

% Plotting OCC
figure('Name', 'Open Circuit Characteristics', 'NumberTitle', 'off');
plot(If, Eg, 'b-', 'LineWidth', 2);
title('Open Circuit Characteristics (OCC)');
xlabel('Field Current, I_f (Amperes)');
ylabel('Generated EMF, E_g (Volts)');
grid on;
hold on;
% Mark the saturation knee point
plot(2.5, (a*2.5)/(b+2.5) + residual_voltage, 'ko', 'MarkerFaceColor',
'k');
text(2.6, 180, 'Magnetic Saturation Begins');

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%% Part 2: Load Characteristics (Separately Excited)
% Simulating Terminal Voltage vs. Load Current at a constant field current

IL = 0:1:50; % Load current (A) ranging from 0 to 50 Amps

% Generator Constants
Ra = 0.8; % Armature resistance (Ohms)
Armature_Reaction_Drop = 0.15; % Voltage drop per Amp due to armature
reaction

% Assume the generator is operating at a constant rated field current
(e.g., If = 3A)
If_rated = 1.5;
Eg_rated = (a * If_rated) / (b + If_rated) + residual_voltage;

% Calculate Terminal Voltage (V = Eg - IaRa - Armature Reaction)
% For a separately excited generator, Ia = IL
V_terminal = Eg_rated - (IL .* Ra) - (IL .* Armature_Reaction_Drop);

% Plotting Load Characteristics
figure('Name', 'Load Characteristics', 'NumberTitle', 'off');
plot(IL, V_terminal, 'r-', 'LineWidth', 2);
hold on;

% Plot the ideal no-load voltage for comparison
plot(IL, Eg_rated * ones(1, length(IL)), 'k--', 'LineWidth', 1.5);

title('External Load Characteristics');
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xlabel('Load Current, IL (Amperes)');
ylabel('Terminal Voltage, V (Volts)');
legend('Terminal Voltage (Actual)', 'No-Load Voltage (Ideal)',
'Location', 'southwest');
grid on;

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